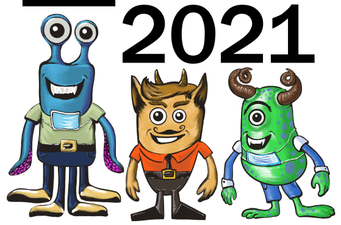
INT4

# FlashPipe

## Simplifying CI/CD for SAP Integration Suite

PIMON 2021





# Agenda

- About Me
- What is Continuous Integration and Continuous Delivery/Deployment?
- Introduction to FlashPipe
- Demo of using FlashPipe with SAP Integration Suite
- Questions & Answers



# About Me



/in/engsweeyeoh

## Eng Swee Yeoh

people.sap.com/engswee.yeoh

**Integration Architect at Int4.**

Integrating SAP solutions since 2005. Working with SAP Integration Suite since 2015.

Author of multiple open-source solutions for SAP integration.

SAP Mentor, TechEd Speaker, SAP PRESS Author



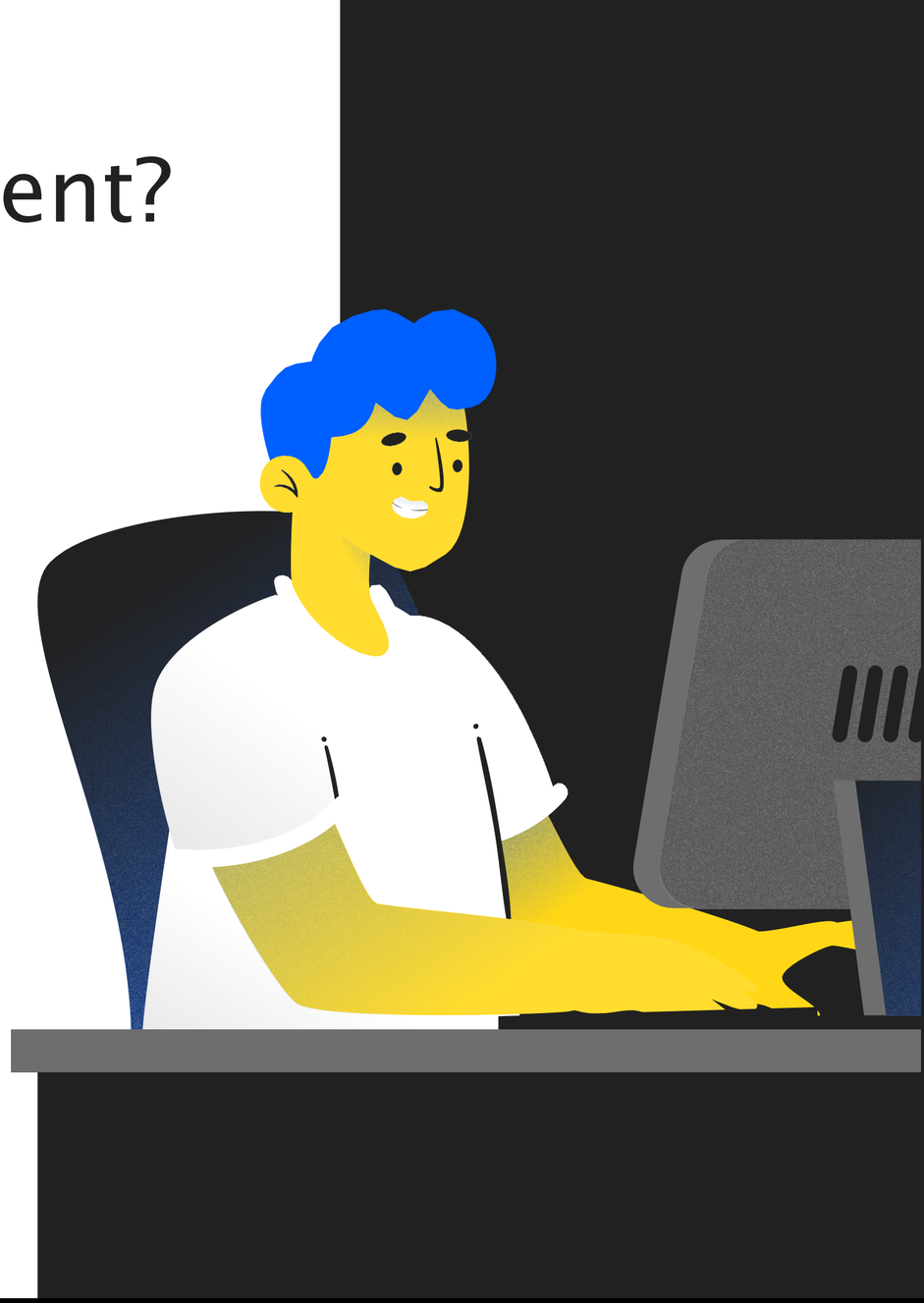

deploys code to production every

11.7 seconds

# What is Continuous Integration and Continuous Delivery/Deployment?

*The ability for rapid deployment to production is backed by automation*

- **Continuous Integration (CI) and Continuous Delivery or Continuous Deployment (CD) enables automation of building, testing and deployment of software**
- **CI/CD capability is now a must-have for IT organizations across all industries**



Point to ponder:
**How long does it take for deployment of an average SAP-based development - 1 week, 1 month, 6 weeks?**

# A Look at the Build-to-Deploy Process for Cloud Integration

**The Build-to-Deploy process of software development in SAP ecosystem consists of many manual and time-consuming steps – it is no different for Cloud Integration**

BUILD

- **Develop Integration Flow in Web UI**
- **Develop Groovy scripts in external IDE (optional)**
- **Manual deployment of design artifact to runtime**

TEST

- **Manual testing using tools or with third party systems**

DEPLOY

- **Manual export of design artifact from development system**
- **Manual import of design artifact into production system**
- **Manual configuration and deployment to runtime**



# Introduction to FlashPipe

***Key features of FlashPipe:***

- **Automated** syncing of Integration Flows to Git repository
- **Automated** unit testing & simulation testing with Maven
- **Automated** deployment of Integration Flows to runtime
- **Automated** deployment to QA, Production environments

**Enables automation across multiple personas/teams – development, testing/QA, change & release**

***Open source (https://github.com/engswee/flashpipe) and licensed under Apache License Version 2.0***



# Key Technologies











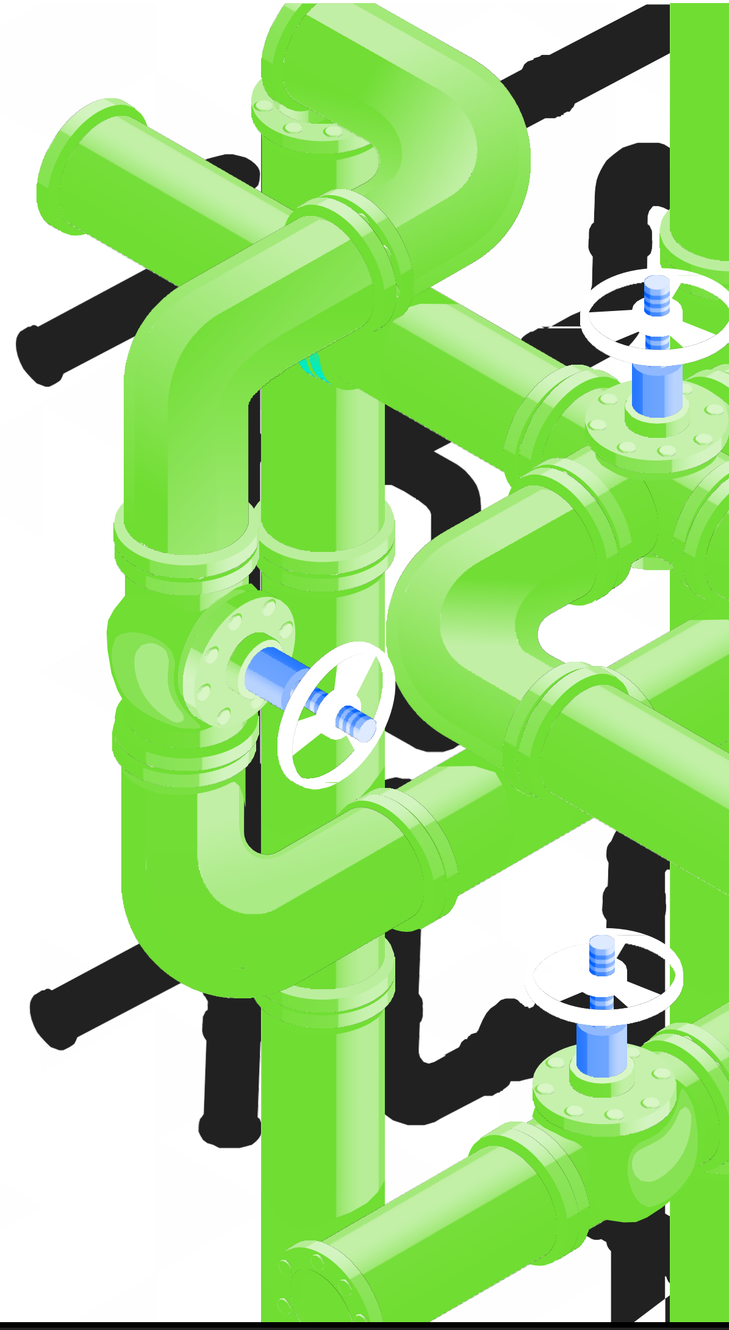

# Closer Look at FlashPipe

**FlashPipe is a public Docker image that provides CI/CD capabilities for SAP Integration Suite**

***It is technically possible to use FlashPipe on any CI/CD tool that supports container-based pipeline execution.***

**Images comes in two flavors (i.e., tags), and can be mixed-and-matched in different steps/stages of a pipeline.**

| Full (e.g., tag 2.2.1) | Lite (e.g., tag 2.2.1-lite) |
| --- | --- |
| Full image with Maven capabilities and selected third-party libraries | Smaller image without Maven. Faster pull time from Docker Hub. |



# Closer Look at FlashPipe

**FlashPipe comprises of the following key components:**

## Java executables

- **Groovy source codes (compiled as Java binaries) for accessing SAP Integration Suite APIs**

## Unix scripts

- **BASH scripts wrapping around Java executables, providing executable steps used in CI/CD pipeline definition**

## Local Maven repository (only in full images)

- **Provides cached libraries for faster execution of Maven-based testing**



# FlashPipe in Action

1. **Create new Integration Flow with dummy Groovy script**
2. **Sync Integration Flow to Git repository with GitHub Actions**
3. **Develop Groovy script in IntelliJ IDEA**
4. **Update and deploy Integration Flow with Azure Pipelines**



# Open questions, feedback session

# Thank you!



**Eng Swee Yeoh**
Integration Architect
SAP Mentor

**www.int4.com**

INT4 IFTT

SAP Silver Partner

SAP® Certified
Integration with SAP® S/4HANA

SAP® Certified
Powered by SAP NetWeaver®